

Homework Assignment 3

1. At a certain location in a flow, the stagnation conditions are 300 kPa and 700 K, the Mach number is 0.8 and the flow area is 0.05 m^2 . $\gamma = 1.3$, $R = 277 \text{ J/kg-K}$.
(a) Without calculating the static conditions, find the mass flow rate. {Ans.: 21.87 kg/s}
(b) Calculate the value of A^* from the result of (a). {Ans.: 0.0481 m^2 }
(c) Now, find the static conditions as well as the velocity at the location. From these values find the mass flow rate.
2. Consider the isentropic flow of a gas ($\gamma = 1.3$) through a c-d nozzle. The Mach number at entry is 0.5. Given that the flow areas at the entry, the throat and the exit are 0.07 m^2 , 0.055 m^2 and 0.08 m^2 , respectively, find the throat and exit Mach numbers. **Fully** (italicized, underlined, and in bold font) explain how you arrived at the results. You may use the compressible aerodynamics calculator in solving this problem.
3. If the isentropic flow in the nozzle of Problem 2 is known to be choked, calculate the entry and the exit Mach numbers, assuming (a) entry flow is subsonic and exit flow is supersonic, and (b) entry flow is supersonic and exit flow is subsonic. You may use the compressible aerodynamics calculator in solving this problem. {Ans.: (a) 0.5427, 1.7757, (b) 1.6016, 0.4516}
4. What is an overexpanded jet? underexpanded jet?
5. Show the variation of (A/A^*) with Mach number in a sketch and clearly identify the subsonic and supersonic regions in it.